

Team JANUS Problem Statement 2026-2027

Subsystem: Avionics

FINAL ROUND

Common Instructions:

1. First off, congratulations on making it this far! We hope that this has been a thrilling experience where you've learnt a lot. We just wanted to take this moment to let you know that no matter what happens here today, the things you've learnt will help you not only in future inductions, but also help you expand your knowledge as well.
2. Alright, the gloves are off now; there are no cheeky references here. Time for you to lock in. You have been given 3 questions in this document. It's up to you to choose to solve one question or more than one of them. It is *not* mandatory to complete (or even attempt) all 3 questions, but if you want to show us that you're able to get more work done and multitask, be our guest. It is fine to present incomplete solutions, but the final presentation is completely in your hands. With that being said, we value 1 complete solution with the second being partial or non-existent more than 2 partially completed solutions.
3. All the files have to be uploaded to a GitHub repository and the link to that repository has to be uploaded to a Google Form. We will give you the link for the 15 minutes before your time is over.
4. This is a true free-for-all; all tools are allowed! You are free to use the Internet and AI if you want. However, we don't want someone who just vomits whatever AI says back to us. We need you to be able to justify your code and know the ins and outs of your answer. As a rule of thumb, use AI as a tool, not the solution.
5. Questions have *not* been ordered according to difficulty. You are free to attempt them in any order. Have fun! (you're specifically requested to)

Question 1 (Ground Control System):

Using Python, write code to read telemetry from an XBee3 (decode it to a string) and plot the latitude, longitude and altitude live on a 3D graph. The graph doesn't need to be fancy, just functional. Use the Digi XBee library.

Example Telemetry Packet:

"Timestamp(int), State(int), Temperature(float), Pressure(float), Altitude(float), Battery Voltage(float), Battery Current(float), Latitude(float), Longitude(float), Prev_CMD_echo(str)"

Question 2 (Paraglider Control):

Using Python, implement the dynamics model discussed in [this paper](#). Based on this model, implement a path finding algorithm for the paraglider where you spawn at a height of 700m (randomize x and y coordinates) and your target location is 0,0,0. You're free to implement a paper or come up with your own guidance algorithm. Plot the target location and the path followed by the glider. You can submit just pseudocode but we will dock off some points. Bonus points if you can convert the latitude, longitude and altitude data from Question 1 into x, y, z coordinates for your algorithm.

Question 3 (PCB Design):

Using the schematic provided, design a PCB in KiCad with the following requirements: a ESP32 S3 Wroom MCU, LSM6DSO and a USB-C port for power. The schematic file already has the electrical circuit for the PCB ready, you just need to route the connections.

Additional Notes:

- You may need to modify component footprints as required.
- Use separate trace widths for power and signal lines.

- Ensure the bottom layer is a ground plane and use vias as needed.
- Auto-Routing is prohibited

Some additional help:

1. Does every question look intimidating and literally impossible? They're designed to be! Tackle each question in small steps and you'll do well enough. Don't be worried if you're googling stuff to figure out what to do, you're supposed to and are on the right path. Some questions might be impossible to answer without googling stuff and reading datasheets and user manuals!
2. Have an idea but don't know how to code it or you simply just don't have the time? Write some pseudocode in a comment on what your code would aim to do and how so. Leaving comments explaining your thought process also shows us how you've been engaging with the questions.
3. Don't be afraid to ask questions! Feel free to reach out to any of us if you have any questions whatsoever. No question is too stupid and we're here to help; we've been in the same position as you:)
4. Have fun with this! (reminder that you are legally mandated to)